

Anway Pimpalkar

PhD Student and NDSEG Fellow at Harvard University

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EDUCATION

Harvard University <i>Cambridge, MA</i>	2025 –
PhD, Engineering Sciences: Materials Science & Mechanical Engineering	
Secondary Field in Mind, Brain and Behaviour	
Advisor: Katia Bertoldi, PhD	
Johns Hopkins University <i>Baltimore, MD</i>	2023 – 2025
MSE, Biomedical Engineering	
Thesis: Bridging Perception & Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Feedback	
Advisor: Jeremy D. Brown, PhD	
College of Engineering Pune <i>India</i>	2019 – 2023
BTech, Electronics and Telecommunication Engineering	

AWARDS & GRANTS

Prize Fellowship • Harvard University, John A. Paulson School of Engineering and Applied Sciences	2025 – 2031
\$281,796 • 6 yr tuition, 2 yr stipend	
National Defense Science & Engineering Graduate Fellowship • US DOW, Office of Naval Research	2025 – 2028
\$277,484 • 3 yr tuition, 3 yr stipend	
GEM Associate Fellowship • National GEM Consortium	2025 – 2031
Full PhD tuition and fees sponsored by Harvard University	
Best Student Paper Finalist [for C2] • International Conference on Rehabilitation Robotics (ICORR)	2025
NIA Start-Up Challenge Finalist [Article] • National Institute on Aging of the NIH	2024 – 2025
\$10,000 grant + 5-month accelerator: 1:1 mentorship, entrepreneurship and SBIR/STTR funding training	
Best Paper Award [for C1] (₹500) • International Conference on Biomedical Engineering Science & Technology	2023
Mitacs Globalink Research Award (CA \$9,915) • Dalhousie University, Canada	2022
Best Project Award [for O1] (₹2,000) • IEEE National Project Competition, India	2021

RESEARCH EXPERIENCE

Harvard University • <i>Bertoldi Group for Materials and Structures by Design</i> ↗	Cambridge, MA
Doctoral Research Fellow • Advisor: Katia Bertoldi, PhD	Oct 2025 –
• Adaptive device design to augment human cognition, behavior, and performance (with Patrick Slade, PhD ↗).	
Boston Children's Hospital • <i>Laboratories of Cognitive Neuroscience – Arnett Lab</i> ↗	Boston, MA
Visiting Doctoral Researcher • Advisor: Anne Arnett, PhD	Aug 2026 –
• Studying the neural correlates for optimal stimulation in children with ADHD.	
Johns Hopkins University • <i>Haptics and Medical Robotics Lab</i> ↗	Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD	Jan 2024 – Jun 2025
• Multimodal visual-haptic neurorehabilitation for improving post-stroke finger dexterity. [J4 , C2 , C3 , C5 , A6 , A7]	
• Soft multimodal haptic interfaces. [J3 , A3 , A4]	
Harvard University • <i>Biohybrid Organs and Neuroprosthetics Lab</i> ↗	Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD	May – Aug 2024
• Peripheral nerve stimulation to modulate cancer tumor metastasis.	

- Surgical approach for treating stress urinary incontinence.

IIT Bombay • *Human Motor Neurophysiology and Neuromodulation Lab* [↗](#)

Mumbai, India

Research Intern • Advisor: Nivethida Thirugnanasambandam, PhD

Jan – Aug 2023

- Quantification of the prospective component of sense of agency using machine learning.

College of Engineering Pune • *Department of Applied Sciences* [↗](#)

Pune, India

Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD

Aug 2022 – Aug 2023

- Deep learning-based MRI skull tissue segmentation. [[C1](#), [A1](#)]

Dalhousie University • *Agricultural Mechanized Systems Group* [↗](#)

Truro, Canada

Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD

May – Aug 2022

- Automation of industrial wild blueberry harvesters to reduce operator cognitive load.

Queliz Lifetech • *Early Stage Medtech Startup*

Pune, India

Research Intern, Electronics Subsystem

Jun – Sep 2021

- Point-of-care device for finger dexterity rehabilitation in acute burn patients.

JOURNAL PUBLICATIONS

- J4 **Pimpalkar A**, Brown JD. *Soft 3D-printed interface for co-located transient and continuous tactile stimuli*. In prep.
- J3 **Pimpalkar A**, West AM, Rai D, Xu J, Brown JD. *Perceptually calibrated haptic feedback provides limited benefit beyond vision in a virtual precision grip task*. In review. 2026.
- J2 Harris C, **Pimpalkar A**, Aggarwal A, Yang J, Chen X, Schmidgall S, Rapuri S, Greenstein JL, Taylor CO, Stevens RD. *Preoperative risk prediction of major cardiovascular events in noncardiac surgery using the 12-lead electrocardiogram: explainable deep learning approach*. British Journal of Anaesthesia. 2025 Sep; doi: 10.1016/j.bja.2025.07.085 [↗](#)
Featured in *The Hub at Johns Hopkins* [↗](#)
- J1 **Pimpalkar A**, Slepian A, Thakor NV. *Vibrations at first contact encode object stiffness before grasp completion*. IEEE Sensors Letters. 2025 Aug. doi: 10.1109/lsens.2025.3584860 [↗](#)
Featured in *New Scientist Magazine* [↗](#)

CONFERENCE PRESENTATIONS [†](#) presenter

FULL-LENGTH PAPERS

- C5 West AM, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectional: Implications for human multisensory feedback*. IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2026 Aug. In print. [Talk]
- C4 Salgado AS, Bartlett JS, **Pimpalkar A**, Brown JD, Reed KB [†](#). *Teaching at the Teachable Moment Using Asynchronous Videos Combined with Live Interactions*. ASEE Annual Conference & Exposition. 2026 Jun. doi: 10.18260/1-2--60737 [↗](#) [Talk]
- C3 West AM [†](#), **Pimpalkar A**, Xu J, Brown JD. *SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments*. International IEEE/EMBS Conference on Neural Engineering (NER), 2025 Nov. p. 85-90. doi: 10.1109/ner61569.2025.11589033 [↗](#) [Spotlight Poster]
- C2 **Pimpalkar A** [†](#), West AM, Xu J, Brown JD. *Optimizing cross-modal matching for multimodal motor rehabilitation*. International Conference on Rehabilitation Robotics (ICORR). 2025 May. P. 559-566. doi: 10.1109/icorr66766.2025.11063112 [↗](#) [Talk] **Best Student Paper Award Finalist**.
- C1 **Pimpalkar A** [†](#), Patole RK, Kamble KD, Shindikar MH. *Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans*. International Conference on Biomedical Engineering Science and Technology. 2023 Feb. p. 131-142. doi: 10.1007/978-3-031-54547-4_11 [↗](#) [Talk] **Best Paper Award**.

ABSTRACTS AND SHORT PAPERS

- A7 West AM [†](#), Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectionally invertible: Implications for multisensory representation*. Neural Control of Movement Meet. 2026 Apr. [Poster]

- A6 **Pimpalkar A** †, Rai D, Bartels UJ, Xu J, Brown JD. *Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task*. Biomedical Engineering Society Annual Meeting (BMES). 2024 Oct. [Talk]
- A5 Song H † et al. [including **Pimpalkar A**]. *Towards visual function restoration through photoacoustic stimulation*. Investigative Ophthalmology & Visual Science ARVO Annual Meeting. 2024 Jun. [Talk]
- A4 **Pimpalkar A** †, Ameta P †, Dalia A, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback*. IEEE Haptics Symposium. 2024 Apr. [Live Demonstration, Poster]
- A3 **Pimpalkar A** †, Ameta P †, Dalia A †, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback*. US Senate AI Caucus Robotics Demo Day. 2024 Apr. [Live Demonstration]
- A2 Harris C †, **Pimpalkar A**, Aggarwal A, Yang P, Chen X, Taylor CO, Greenstein J, Stevens RD. *Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms*. Johns Hopkins Medicine & Engineering Research Retreat. 2024 Feb. [Poster]
- A1 **Pimpalkar A** †, Patole R, Kamble K, Shindikar M. *Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans*. No Garland Neuroscience Conference. 2023 Feb. [Podium]

NON PEER-REVIEWED ARTICLES

- O1 **Pimpalkar A**, Niture D. *Towards contactless elevators with tinyML using person detection and keyword spotting*. 2024 Jul. doi: arXiv:2405.13051 [↗](#) **Best Project Award**.

INVENTIONS DISCLOSED

- P1 *Deep Learning Models for Predicting Major Adverse Cardiovascular Events in Noncardiac Surgery*. 2026 Aug. Disclosed to Johns Hopkins Technology Ventures.

TEACHING EXPERIENCE

Engineering Design • Harvard University ES 100 Fall 2026, Spring 2027

Teaching Fellow to Eric Richardson, PhD

- Mentoring 6 student teams through year-long, industry-client-proposed capstone projects using an agile design methodology, spanning robotics and engineering applications in human-robot interaction, lightweight structures, sustainability, complex-environment navigation, and surgical robotics.

Haptic Interface Design for Human-Robot Interaction • Johns Hopkins University 530.691 Fall 2024

Class Researcher with Jeremy D. Brown, PhD

- Assisted the lead instructor in piloting an interactive video platform that anchors student questions and recorded instructor responses to lecture timestamps, bringing live-instructor interaction into an otherwise self-paced hardware design curriculum; co-authored resulting ASEE 2026 paper. [\[C4\]](#)

Principles of the Design of Biomedical Instrumentation • Johns Hopkins University 580.771 Fall 2024

Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG circuits and 24 projects ranging accessible gaming and wearables.

Mechatronics • Johns Hopkins University 530.421 Spring 2024, 2025

Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge of enabling real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking. Led labs on building autonomous robots. [Article ↗](#) • [GitHub ↗](#)

ENTREPRENEURIAL EXPERIENCE

Re-Kinesis • CEO and Co-Founder • NIA Start-Up Challenge Finalist Baltimore, MD

[JH Hub Article ↗](#) • [NIH NIA Announcement ↗](#) Nov 2024 – Sep 2025

- Co-founded and led an early-stage medical device venture developing a high-resolution, high-speed insole tactile sensor that reconstructs full gait kinematics from plantar force data alone, enabling continuous mobility assessment in aging populations without motion-capture infrastructure.

MENTORING

UNDERGRADUATES

Jocelyn Hwang • Summer Intern from MIT, BS Electrical Engineering and Computer Science Project: Multiplexing electronics for multi-electrode biosensing.	Jun – Aug 2026
George Sealy • Summer Intern from University of Cambridge, BA/MSci Materials Science Project: Mechanics of elastic structures.	Jul – Aug 2026
Fu Xing Chen • Undergraduate Researcher from MIT, BS Mechanical Engineering Project: Mechanics and characterization of viscoelastic actuators.	Jan – Jul 2026

INVITED PRESENTATIONS AND LECTURES

T2 Indian Institute of Technology Bombay • Human Motor Neurophysiology & Neuromodulation Lab	Jun 2025
T1 Johns Hopkins University • Neural & Rehabilitation Engineering Course	Sep 2024

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer	2024
Canadian Society of Bioengineering Annual Meeting • Student Volunteer	2022

IN MEDIA

Johns Hopkins University Hub ↗ “AI fares better than doctors at predicting deadly complications after surgery”	Sep 17, 2025
New Scientist Magazine ↗ “Delicate robot hands know just how hard to squeeze”	Jan 2, 2025
Johns Hopkins University Hub ↗ “A game-changing gait lab in a shoe”	May 9, 2025
Johns Hopkins University Hub ↗ “Robotics teams shoot it out in JHockey tournament”	May 16, 2024
Arduino Blog ↗ “This contactless system combines embedded ML and sensors to improve elevator safety”	Jan 29, 2022

LEADERSHIP AND TEAMWORK

Johns Hopkins Gupta-Klinsky India Institute • Operations Team • Managed on-day operations and logistics for the first annual Hopkins-India Conference.	2024 – 2025
Johns Hopkins Biomedical Engineering • Master's Program Representative • Represented the department at information sessions and conferences, including BMES 2024, upon invitation.	2024 – 2025
COEP Impressions Cultural Festival • Head of Events (2021), Volunteer • Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinating with an overall team of 126 members to execute concerts, competitions, and celebrity engagements. • Built and directly led a team of 15, working with 2 co-heads to build a platform from the ground up.	2019 – 2021
Rowing Team • COEP, Pune City • 1X, 2X, 4X Sculler • Won race events and an institutional award commending skill and spirit. Represented Pune City at regionals.	2020 – 2022

REFERENCES

Katia Bertoldi, PhD • Doctoral Advisor

William and Ami Kuan Danoff Professor of Applied Mechanics
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Harvard University
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